

Challenges Faced

One of the main challenges was ensuring the data format matched what TensorFlow Lite expects during inference. When running inference, the model input requires the correct shape and data type (typically float32). A small mistake like missing the batch dimension (using shape 28×28 instead of $1 \times 28 \times 28$) can cause inference errors or unexpected output. Another challenge was understanding the difference between training a model in TensorFlow/Keras and running inference with TensorFlow Lite, because TensorFlow Lite uses a more manual workflow (setting tensors and invoking the interpreter) compared to the simplicity of `model.predict()` in Keras.

What I Learned

This lab helped me understand the full pipeline from model training to deployment. I learned that training a model is only part of what is needed in real applications—deployment requires converting models into formats that are efficient and compatible with the target device. I also learned the importance of preprocessing consistency: the same normalization used during training needs to be applied during inference, or the model's predictions can become less accurate. Additionally, I gained exposure to how TensorFlow Lite works internally, including interpreter tensor allocation, input/output tensor details, and invoking inference step-by-step.

How TensorFlow Lite Applies to Real-World AI Deployments

TensorFlow Lite is useful for deploying AI models on devices with limited compute, memory, and power—such as smartphones, microcontrollers, and IoT sensors. In real-world scenarios,

deploying models on edge devices has important benefits: faster response time (low latency), reduced dependence on internet connectivity, improved privacy (data processed locally), and lower cloud costs. For example, handwritten digit recognition is similar in concept to real edge applications like on-device OCR, mobile camera recognition, and smart sensor classification. This lab showed me how a model can be trained in a normal environment and then converted into a lightweight version that is practical for deployment, which is a key step in production AI systems.

Final Thoughts

Overall, this lab clarified that “building AI” is not only about getting a good accuracy score in training—it also includes making the model usable in practical environments. TensorFlow Lite is a balance between model development and real deployment, and understanding its workflow is essential for real-world AI engineering.